Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Ethernet client port

File: eth_client.kicad_sch

Ethernet I/O port

File: eth_io.kicad_sch

Memories and display

File: periph.kicad_sch

Field digital I/O

File: io_dig.kicad_sch

Field power, pump, relays, USB

File: io_pwr.kicad_sch

Analog inputs and temperatures

File: io_ana.kicad_sch

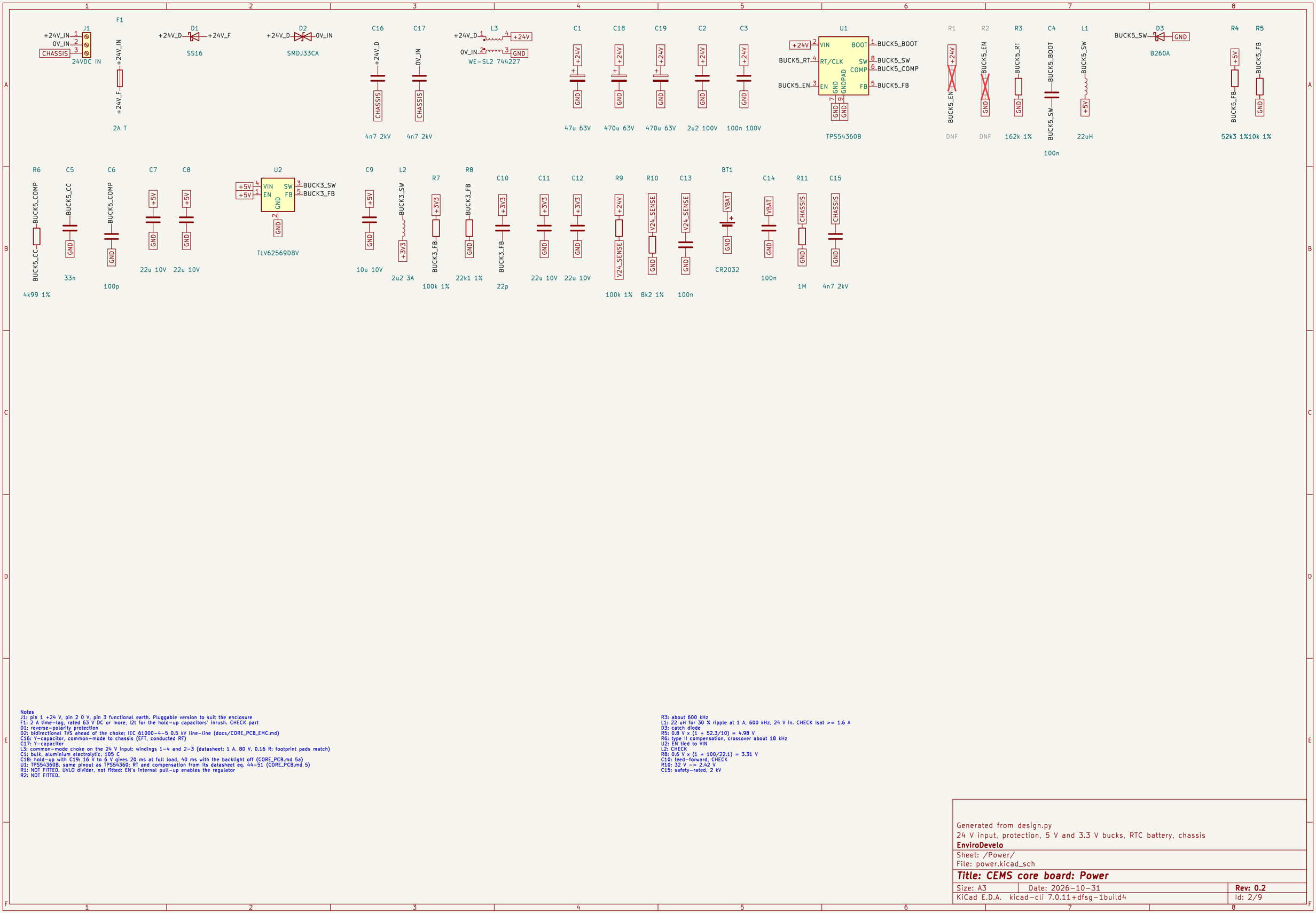
CEMS core board, rev 0.1 schematic (draft for review)
Two separate Ethernet networks: client port = STM32H573 MAC + LAN8742A (J2); I/O port = W5500 (J3). I/O boards only.
24 VDC in; DIN-rail enclosure behind the RVT70 display (FFC J4).
Pin map and design notes: docs/CORE_PCB.md. Parts marked CHECK in the sheet notes need datasheet confirmation before layout.

Design spec: docs/CORE_PCB.md
Generated from hardware/core_pcb/tools/design.py; do not edit by hand
EnviroDevelo

Sheet: /
File: cems_core.kicad_sch

Title: CEMS core board

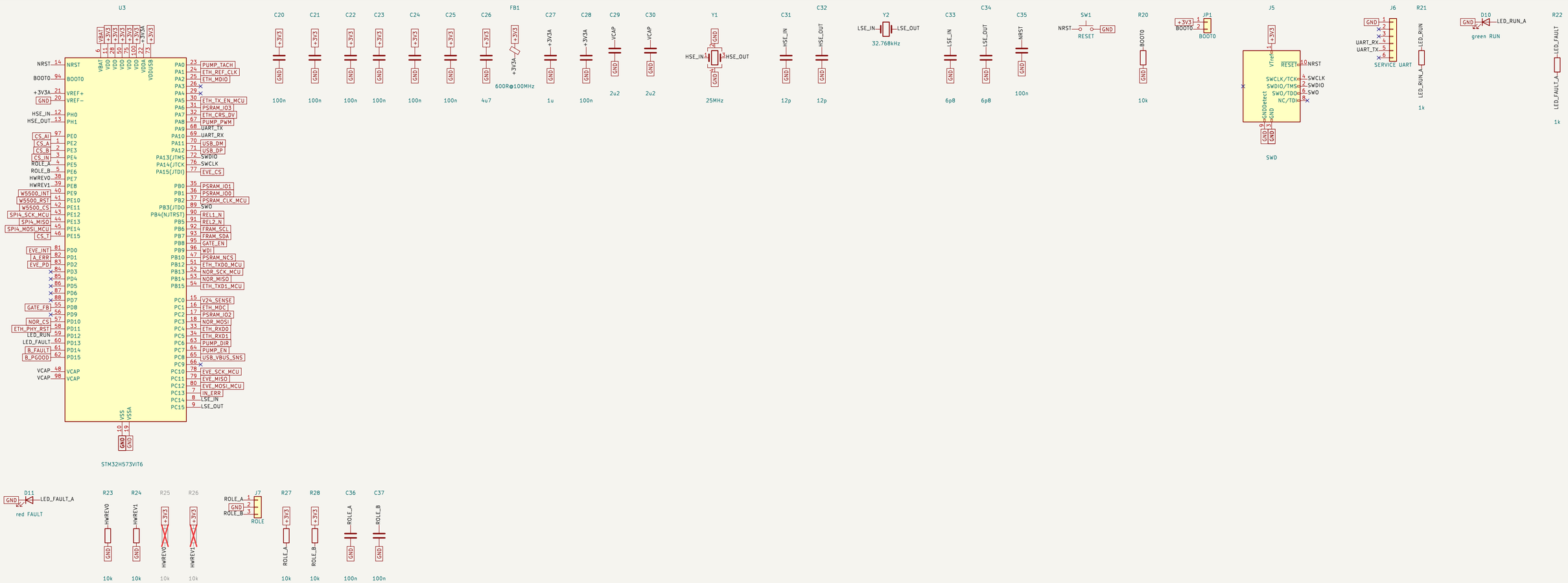
Size: A4	Date: 2026-10-31	**Rev: 0.2**
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Notes
J1: pin 1 +24 V, pin 2 0 V, pin 3 functional earth. Pluggable version to suit the enclosure
F1: 2 A time-lag, rated 63 V DC or more, I2t for the hold-up capacitors' inrush. CHECK part
D1: reverse-polarity protection
D2: bidirectional TVS ahead of the choke; IEC 61000-4-5 0.5 kV line-line (docs/CORE_PCB_EMC.md)
C16: Y-capacitor, common-mode to chassis (EFT, conducted RF)
C17: Y-capacitor
L3: common-mode choke on the 24 V input; windings 1-4 and 2-3 (datasheet: 1 A, 80 V, 0.16 R; footprint pads match)
C1: bulk, aluminium electrolytic, 105 C
C18: hold-up with C19: 16 V to 6 V gives 20 ms at full load, 40 ms with the backlight off (CORE_PCB.md 5a)
U1: TPS54360B, same pinout as TPS54360; RT and compensation from its datasheet eq. 44-51 (CORE_PCB.md 5)
R1: NOT FITTED. UVLO divider, not fitted: EN's internal pull-up enables the regulator
R2: NOT FITTED.
R3: about 600 kHz
L1: 22 uH for 30 % ripple at 1 A, 600 kHz, 24 V in. CHECK Isat >= 1.6 A
D3: catch diode
R5: 0.8 V x (1 + 52.3/10) = 4.98 V
R6: type II compensation, crossover about 18 kHz
U2: EN tied to VIN
L2: CHECK
R8: 0.6 V x (1 + 100/22.1) = 3.31 V
C10: feed-forward, CHECK
R10: 32 V -> 2.42 V
C15: safety-rated, 2 kV

Generated from design.py
24 V input, protection, 5 V and 3.3 V bucks, RTC battery, chassis
EnviroDevelo

Sheet: /Power/		
File: power.kicad_sch		
Title: CEMS core board: Power		
Size: A3	Date: 2026-10-31	**Rev: 0.2**
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Notes
U3: STM32H563VIT6 is pin-compatible
C20: one per VDD pin, VDDUSB included
C21: one per VDD pin, VDDUSB included
C22: one per VDD pin, VDDUSB included
C23: one per VDD pin, VDDUSB included
C24: one per VDD pin, VDDUSB included
C25: one per VDD pin, VDDUSB included

FB1: VDDA / VREF+ filter
C29: VCAP, CHECK value and count against the H5 datasheet
Y1: HSE, CL 10 pF class, CHECK against the DK
C31: CHECK for the crystal's CL
Y2: CL 6 pF, ESR per AN2867
C33: AN2867, CHECK for the crystal
JP1: fit a jumper for the ST bootloader

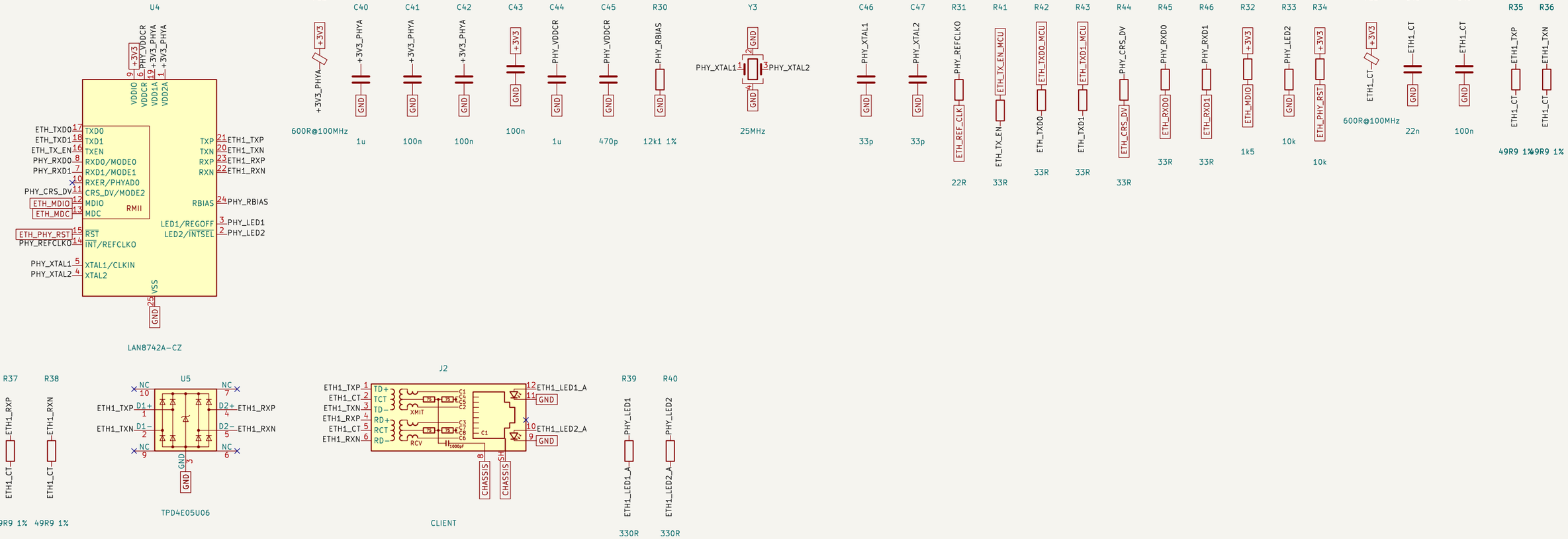
J6: FTDI TTL-232R-3V3 order: 1 GND, 2 CTS, 3 VCC (not connected), 4 cable TXD, 5 cable RXD, 6 RTS
R23: hardware revision 0: both pulled down
R25: NOT FITTED.
R26: NOT FITTED.
J7: shunt with pull tab; silkscreen 1–2 CORE / 2–3 I/O NODE / none = OFF: reachable only with the housing open
C36: ESD, at J7

Generated from design.py
STM32H573VIT6, clocks, reset, boot, SWD, service UART, LEDs
EnviroDevelo

Sheet: /MCU/
File: mcu.kicad_sch

Title: CEMS core board: MCU

Size: A2	Date: 2026-10-31	**Rev: 0.2**
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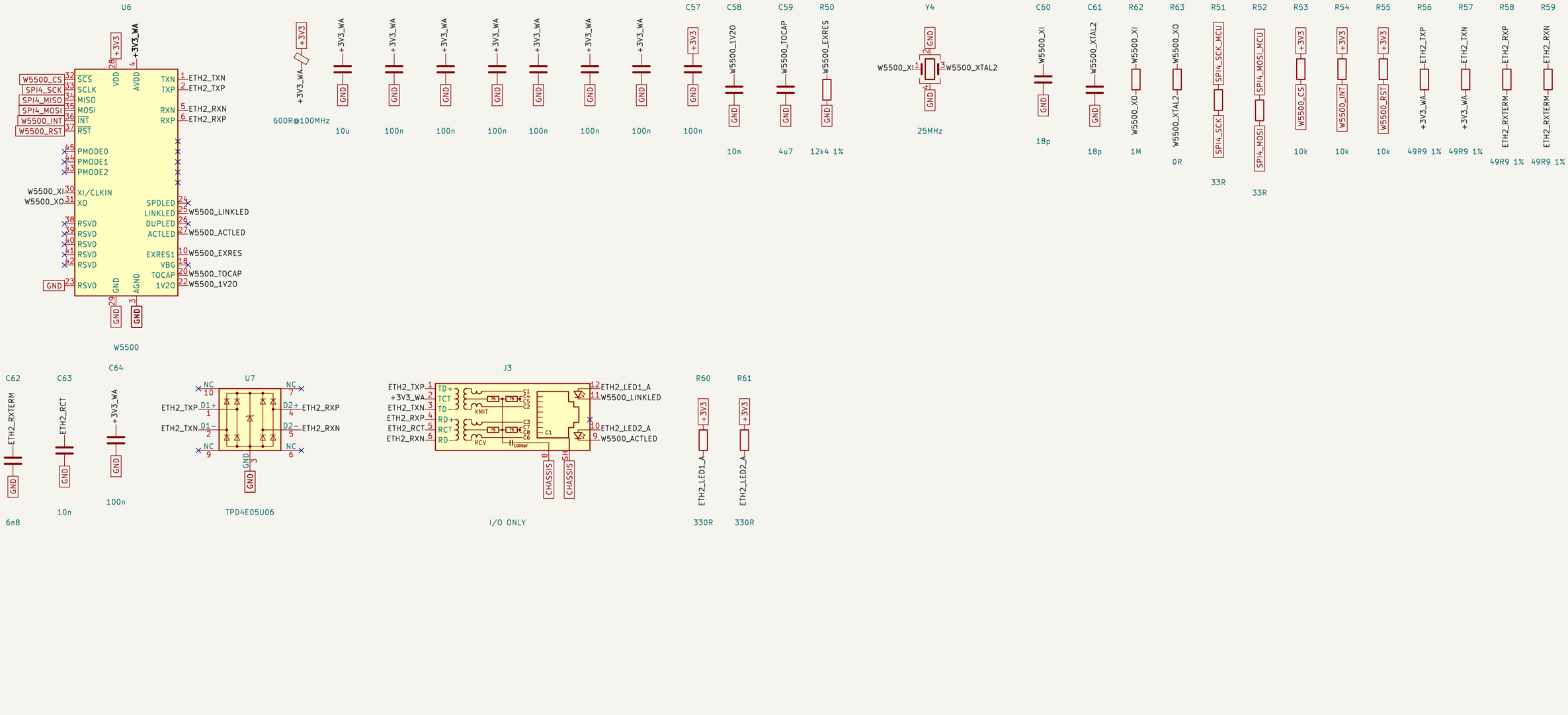
Notes
U4: straps: MODE[2:0] internal pull-ups = all capable; PHYAD0 internal pull-down = address 0; nINTSEL pulled low (R33) = pin 14 is REFCLKO, 50 MHz to the MCU; REGOFF internal pull-down = regulator on
C41: at pin 1
C42: at pin 19
C43: VDDIO
C44: VDDCR, with C45
Y3: CL 20 pF, +/-50 ppm total (LAN8742A datasheet)
C46: for CL 20 pF with about 4 pF stray
R31: series, at the PHY
R41: series, at the MCU
R42: series, at the MCU
R43: series, at the MCU
R44: series, at the PHY
R45: series, at the PHY
R46: series, at the PHY
R33: strap nINTSEL = 0 (REFCLKO mode)
R34: PHY runs if the firmware leaves PD11 as input
U5: flow-through: route each line across pins 1/10, 2/9, 4/7, 5/6
J2: client network
R39: LED1 = link/activity, active high (REGOFF strap low)
R40: LED2 = speed, active high (nINTSEL strap low)

Generated from design.py
LAN8742A on RMII, client RJ45 (J2)
EnviroDevelo

Sheet: /Ethernet client port/
File: eth_client.kicad_sch

Title: CEMS core board: Ethernet client port

Size: A3	Date: 2026-10-31	**Rev: 0.2**
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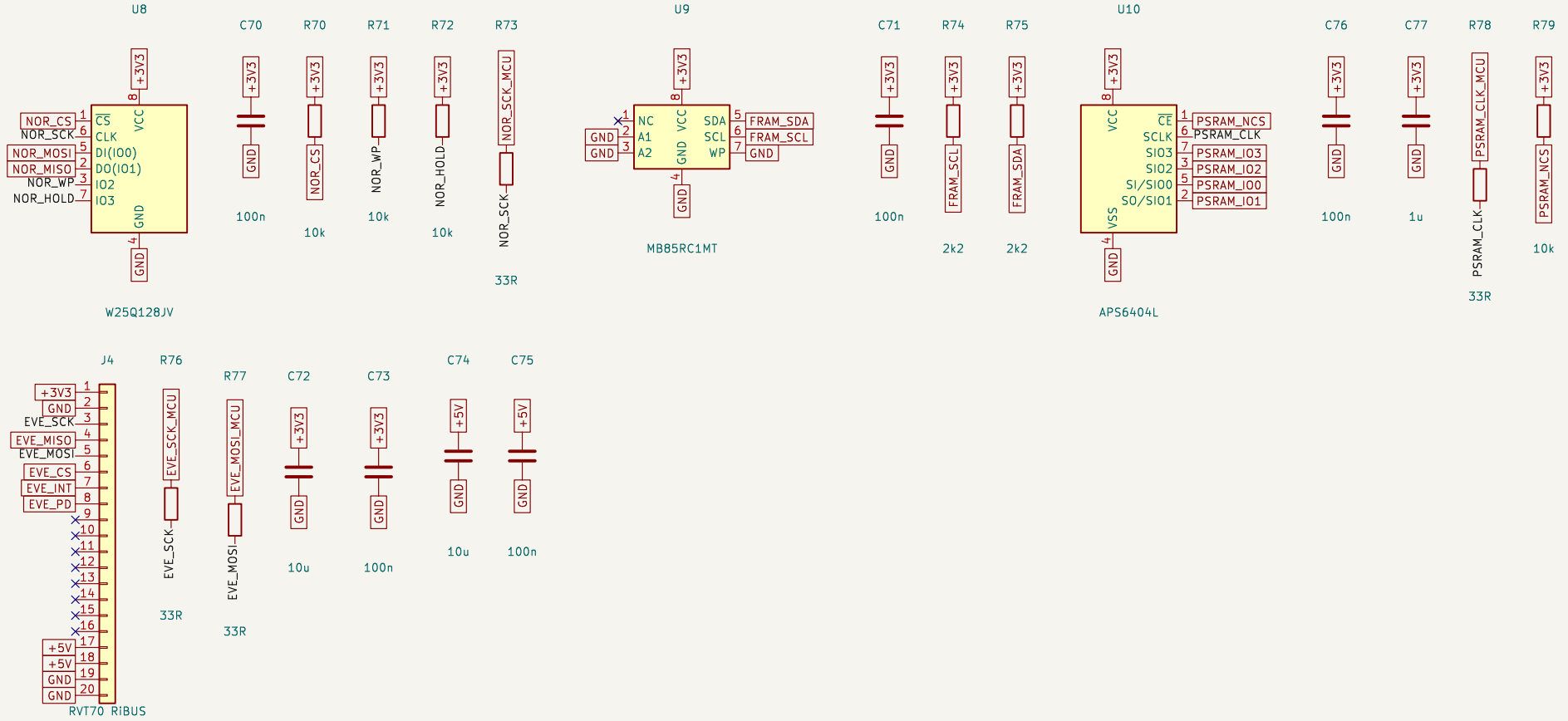


Notes
U6: PMODE[2:0] internal pull-ups = all capable, auto-negotiation; RSVD pin 23 to ground, RSVD 38–42 NC; VBG open. No auto-MDIX: the I/O board's LAN8742A does HP auto-MDIX, so a straight cable works
C51: one per AVDD pin
C52: one per AVDD pin
C53: one per AVDD pin
C54: one per AVDD pin
C55: one per AVDD pin
C56: one per AVDD pin
C57: VDD pin 28
Y4: own crystal, CL 18 pF, +/-30 ppm (W5500 datasheet); separate from the MCU's HSE

C60: W5500 reference circuit
R62: W5500 reference circuit
R63: drive-level option, W5500 reference circuit
R51: series, at the MCU
R52: series, at the MCU
C64: TX centre tap
J3: I/O boards only; silkscreen I/O ONLY
R60: link LED; W5500 LED outputs sink (active low)
R61: activity LED

Generated from design.py
W5500 on SPI4, I/O RJ45 (J3), I/O boards only
EnviroDevelo

Sheet: /Ethernet I/O port/		
File: eth_io.kicad_sch		
Title: CEMS core board: Ethernet I/O port		
Size: A3	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/9



Notes
U8: event log and data logger (H3)
R73: series, at the MCU
U9: 128 KB FRAM (H2), I2C address 0x50. A1 = A2 = 0, WP low = writes enabled (pinout per datasheet)
U10: 8 MB QSPI PSRAM on OCTOSPI1, up to about 100 MHz: keep it next to the MCU, traces under 30 mm and roughly matched
C76: at U10 VDD
R78: series, at the MCU
R79: keeps the PSRAM deselected through reset
J4: RiBUS pinout from the RVT70HSBNWC00 V1.1A datasheet; 9–12 (GPIO, audio, QSPI) left open; 13–16 NC. CHECK contact side against the cable at

mechanical layout
R76: series, at the MCU
R77: series, at the MCU
C72: at J4
C73: at J4
C74: at J4, backlight
C75: at J4

Generated from design.py
NOR flash, FRAM, RVT70 display FFC (J4)
EnviroDevelo

Sheet: /Memories and display/
File: periph.kicad_sch

Title: CEMS core board: Memories and display

Size: A4	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/9

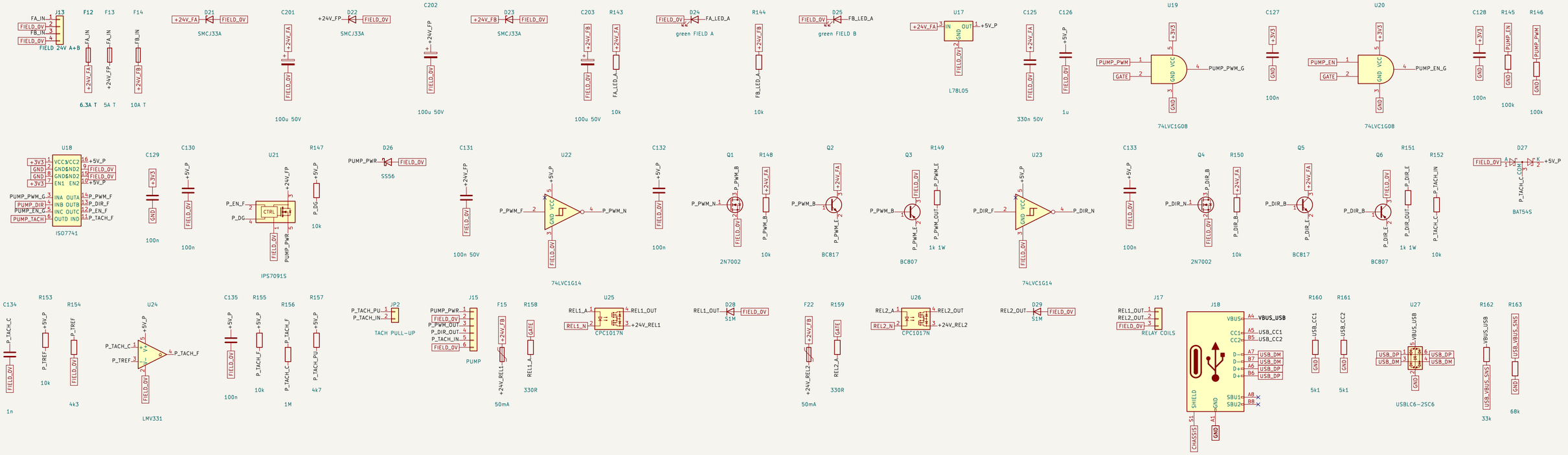
U11
MAX6369
C101
100n
R101
10k
R102
100k
U13
74LVC1G175
C102
100n
U12
74LVC1G08
C103
100n
R103
1k
U14
ISO2H823V2.5
C104
2u2
C105
100n
R104
10k 0.1%
R105
6k81 1%
R106
12k1 1%
C106
1u 50V
C107
470n
C108
2u2 50V
C109
100n 50V
R107
10k
C110
12n 100V
C111
12n 100V
C112
12n 100V
C113
12n 100V
C114
12n 100V
C115
12n 100V
C116
12n 100V
C117
12n 100V
U15
ISO808A-1
C118
100n
C119
1u
C120
100n 50V
C200
100u 50V
R108
10k
R109
10k
J11
BANK B
D31
S1M
R110
0R
D41
24V 1.5W
D32
S1M
R111
0R
D42
24V 1.5W
D33
S1M
R112
0R
D43
24V 1.5W
D34
S1M
R113
0R
D44
24V 1.5W
D35
S1M
R114
0R
D45
24V 1.5W
D36
S1M
R115
0R
D46
24V 1.5W
D37
S1M
R116
0R
D47
24V 1.5W
D38
S1M
R117
0R
D48
24V 1.5W
U16
ISO1B13T
C121
100n
C122
10u
C123
470n 50V
C124
470n 50V
R118
22k1 1%
R119
33R
R120
33k
R121
10k
R122
12k 1%
R123
2k 1%
R124
12k 1%
R125
2k 1%
R126
12k 1%
R127
2k 1%
R128
12k 1%
R129
2k 1%
R130
12k 1%
R131
2k 1%
R132
12k 1%
R133
2k 1%
R134
12k 1%
R135
2k 1%
R136
12k 1%
R137
2k 1%
J10
DIGITAL I/O IDC40
F10
0.75A
R138
10k
R139
10k
R140
10k
R141
10k
R142
10k

Notes
U11: SET2..0 = 010: 30 ms startup delay and timeout (minimum), so outputs drop within 50 ms. WDO pulses: U13 latches it
R102: gate shut while the MCU is in reset
U13: D flip-flop; rising GATE_EN sets GATE_ARM, a watchdog trip clears it until the next rising edge
U12: GATE = GATE_EN and armed (no watchdog trip since)
R103: firmware reads the gate back (PDB)
U14: pinout checked against datasheet Rev 2.1 (31 Oct). Serial mode 3: SEL high, MSO/MS1 high, ALE/RST low; ODIS active low from GATE. No LED matrix: X pins tied, Y pins tied. The package has two exposed pads (GND and GNDBB): footprint is a placeholder to redraw from the package drawing
R104: CLKADJ, high-precision 10k (datasheet)
R105: IADJ, 6.81k only (datasheet)
R106: open-load threshold, about 1 mA (2.3k–25k allowed: 5–0.5 mA)
C110: per output, datasheet application circuit

U15: pinout checked against DS14170 (31 Oct). Tab = VCC (+24V_FB): copper pour on the field side only
C200: bank B bulk
J11: B A return over 2 contacts: CHECK the plug rating (8 A per contact)
D31: flyback at the terminal, as on the breakout
D41: NOT FITTED, fast-release option (remove the 0R)
D42: NOT FITTED.
D43: NOT FITTED.
D44: NOT FITTED.
D45: NOT FITTED.
D46: NOT FITTED.
D47: NOT FITTED.

D48: NOT FITTED.
U16: serial mode: internal DC/DC off (DC_ENA low), field side from VBB. Values from Infineon's ISO1B13T evaluation board. Pinout checked against the datasheet (31 Oct): serial mode 3 (24-bit, CRC)
R118: 22.1k: about 500 kHz scan (datasheet typical)
R119: type 1 (eval board RTS = 33 ohm; datasheet: type 2 = 33k, type 3 = 330k). CHECK type 1 value
R120: wire-break current, 25–55k range (datasheet curves)
R122: RIN, eval board
R123: RLED (no LED here: the breakout has them)
J10: same pinout as the breakout J1, standard straight-through 40-way ribbon
F10: ribbon +24 V: sensor supply and breakout LEDs

Generated from design.py
Output gate, bank A 8 x 0.6 A (ISO2H823V2.5), bank B 8 x 1 A (ISO808A-1), 8 inputs (ISO1B13T)
EnviroDevelo
Sheet: /Field digital I/O/
File: io_dig.kicad_sch
Title: CEMS core board: Field digital I/O
Size: A2 | Date: 2026-10-31 | **Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 7/9



Notes
J13: one 4-way plug for both field supplies (12 A per contact): A = bank A, inputs, pump (about 10 A); B = bank B and relays (about 8 A). Link A and B in the plug for a single supply
F12: bank A, inputs, ribbons
F13: pump
F14: bank B, relays
D21: unidirectional TVS; surge clamp, and blows the fuse on reverse polarity
U17: pump logic 5 V, field side
U18: pinout checked (31 Oct)

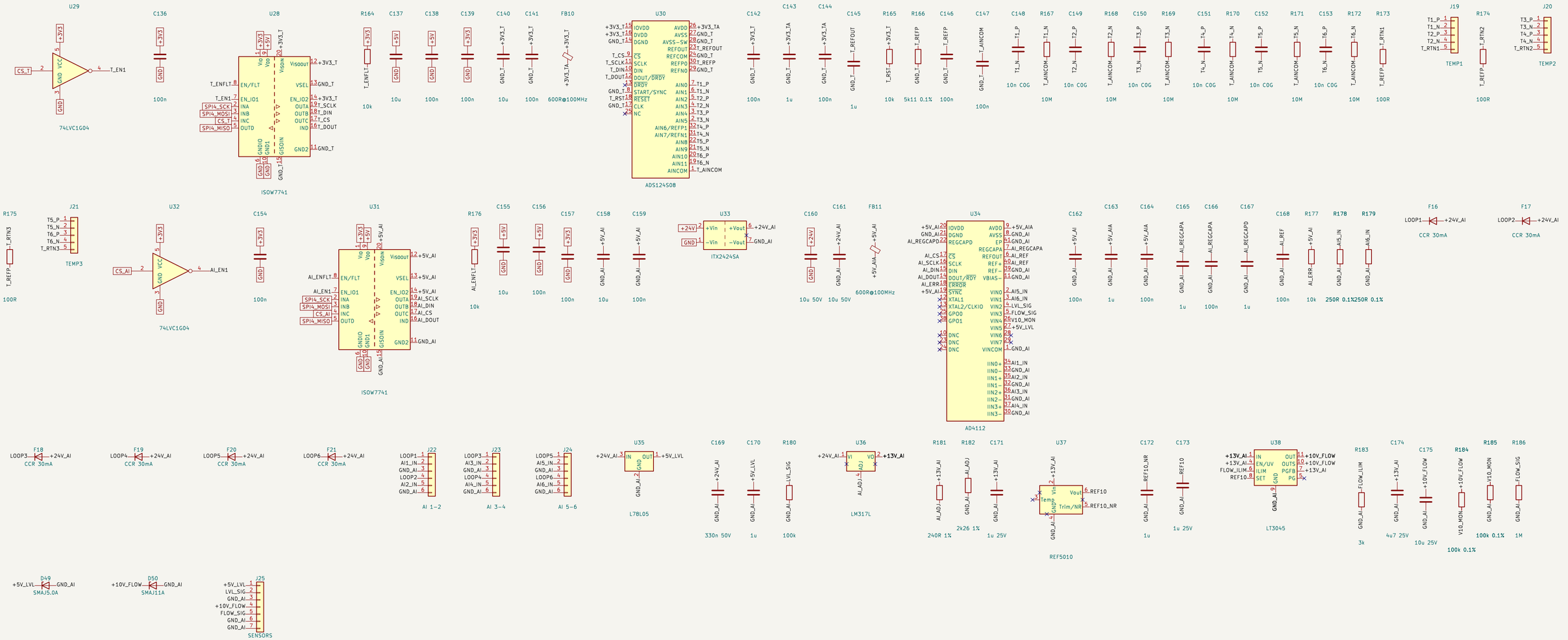
U21: about 4 A pump supply with current limit; can PWM a brushed motor to 1 kHz. CHECK rating once the 107Z current is known
R147: diagnostic, test point only
D26: motor freewheel
R149: pump PWM: 24 V push-pull, 20 Hz–25 kHz, 1k makes the output survive a short to 0 V or 24 V for ever (24 mA, 0.6 W). CHECK the 107Z input current (high level = 24 V − 1k x I)
R151: pump DIR: 24 V push-pull, 20 Hz–25 kHz, 1k makes the output survive a short to 0 V or 24 V for ever (24 mA, 0.6 W). CHECK the 107Z input current (high level = 24 V − 1k x I)
D27: clamp to the 5 V rail
R154: 1.5 V threshold

U24: CHECK pin order for the SOT-23-5 variant
R156: hysteresis
JP2: fit for an open-collector tacho
J15: one plug for the whole pump: power, 0 V, PWM, DIR, tacho, signal 0 V (Thomas 107Z: signal ground must be field 0 V)
F15: per-channel limit: a shorted coil wire trips only its own PTC
U25: 60 V, 400 mA PhotoMOS: rides out a short until its PTC trips. CHECK with a 10 s short test
R163: 5 V -> 3.4 V

Generated from design.py
Field supplies A and B, pump switch / PWM / DIR / tacho, 2 relay coil outputs, USB-C
EnviroDevelo
Sheet: /Field power, pump, relays, USB/
File: io_pwr.kicad_sch

Title: CEMS core board: Field power, pump, relays, USB

Size: A2	Date: 2026-10-31	**Rev: 0.2**
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Notes
U28: VSEL to GND2 = 3.3 V out (140 mA)
R164: EN/FLT high (VIO level) enables the DC/DC; reads low on a fault
U30: pinout checked (SBAS660, 31 Oct). Internal 4.096 MHz clock (CLK to DGND). START low (conversions by command). DOUT/DRDY on one pin
C145: REFOUT to REFCOM
R166: RTD reference resistor, 10 ppm/K
C147: CHECK AINCOM and the TC bias against the Io-node schematic
C148: differential filter
R167: open–TC bias
R173: RTD return to the reference resistor
J19: T1 +/–, T2 +/–. RTD return
J20: T3 +/–, T4 +/–. RTD return

J21: T5 +/–, T6 +/–. RTD return
U31: VSEL to VISOOUT = 5 V out (110 mA)
R176: EN/FLT high (VIO level) enables the DC/DC; reads low on a fault
U33: loop power 6 x 22 mA + level + mass-flow supply, about 4 W
U34: internal oscillator and 2.5 V reference. VIN0/1: 4–20 mA ch 5/6 on shunts; VIN2 level; VIN3 mass flow; VIN4 10 V monitor; VIN5 5 V monitor. Pinout checked (31 Oct); CHECK the exposed pad size against the CP–40 drawing
R178: 4–20 mA shunt -> 1–5 V; takes the 30 mA limit for ever (0.23 W)
R179: 4–20 mA shunt -> 1–5 V; takes the 30 mA limit for ever (0.23 W)
F16: loop current limit; shorted loop = 0.7 W copper on the cathode. CHECK the 30 mA option. Its 2.5 V headroom and the AD4112 IIN abs max
J22: per channel: 24 V loop, IN, 0 V (2–wire or 4–wire transmitters)
J23: per channel: 24 V loop, IN, 0 V (2–wire or 4–wire transmitters)

J24: per channel: 24 V loop, IN, 0 V (2–wire or 4–wire transmitters)
U35: level sensor 5 V, internal current limit
R180: defined level when unplugged
U36: 13 V pre–regulator, about 0.5 W; copper on the VO pins
R182: 1.25 x (1 + 2260/240) = 13.0 V
U38: SET driven by the REF5010 (no RSET). CHECK PGFB tie when PG is unused
R183: about 50 mA current limit
R184: 10 V monitor for ratiometric correction
D49: clamps +5V_LVL
D50: clamps +10V_FLOW (12.2 V min breakdown)
J25: level (DFRobot SEN0205): 5 V, SIG, 0 V; mass flow (AWM5104VN): +10 V, OUT, 0 V

Generated from design.py
Isolated: 6 x 4–20 mA, level and mass–flow ports (AD4112); 6 temperatures (ADS124S08)

EnviroDevelo

Sheet: /Analog inputs and temperatures/
File: io_ana.kicad_sch

Title: CEMS core board: Analog inputs and temperatures

Size: A2	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/9